Problem Set 7: Solution

1. FIR filter classification & frequency response derivation

Given four causal discrete-time FIR systems with impulse responses defined for $0 \leq n \leq M$:

1. $h_1[n] = [1,3,5,3,1]$
2. $h_2[n] = [2,-1,4,4,-1,2]$
3. $h_3[n] = [1,2,0,-2,-1]$
4. $h_4[n] = [-1,3,-3,1]$

(a) Classify each system into its corresponding FIR Type (I, II, III, or IV). Specify the filter order M and sequence length N for each.

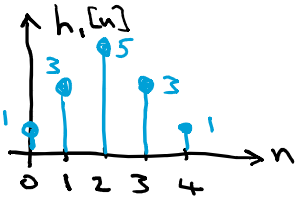
(b) For $h_2[n]$, derive the compact closed-form expression for its frequency response $H_2(e^{j\omega})$. Express $H_2(e^{j\omega})$ in the form $A(\omega)e^{-jM/2}$ where $A(\omega)$ is a real-valued trigonometric sum.

(c) For $h_3[n]$, derive $H_3(e^{j\omega})$ and show that it exhibits a constant 90° ($\pi/2$ rad) phase shift relative to the linear phase delay term $e^{-jM/2}$.

1. $N=5$, $M=4$ even

$h[n] = h[M-n]$ $n = 0,1,2,3,4$

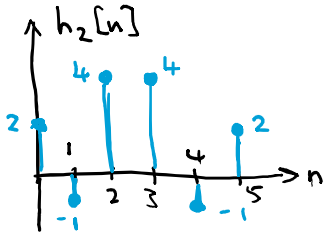
FIR Type I



2. $N=6$, $M=5$ odd

$h[n] = h[M-n]$ $n = 0,1,2,3,4,5$

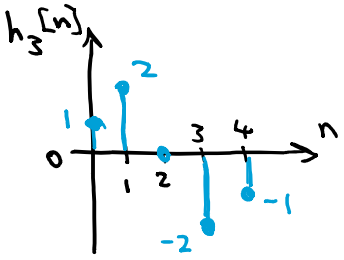
FIR Type II



3. $N=5$, $M=4$ even

$h[n] = -h[M-n]$

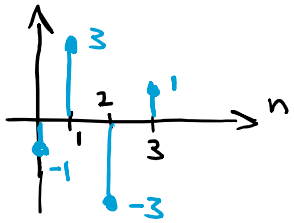
FIR Type III



4. $N=4$, $M=3$ odd

$h[n] = -h[M-n]$

FIR Type IV



(b) $H_2(e^{j\omega}) = 2 - e^{-j\omega} + 4e^{-j2\omega} + 4e^{-j3\omega} - e^{-j4\omega} + 2e^{-j5\omega}$

$= 2(e^{j5\omega/2} + e^{-j5\omega/2})\, e^{-j5\omega/2} +$

$- (e^{j3\omega/2} + e^{-j3\omega/2})\, e^{-j5\omega/2} +$

$+ 4(e^{j\omega/2} + e^{-j\omega/2})\, e^{-j5\omega/2}$

$= \underbrace{\left[4\cos\left(\frac{5\omega}{2}\right) - 2\cos\left(\frac{3\omega}{2}\right) + 8\cos\left(\frac{\omega}{2}\right)\right]}_{=A(\omega)} e^{-j5\omega/2}$

$e^{-j5\omega/2}$: linear phase $-5\omega/2$

(c) $H_3(e^{j\omega}) = 1 - e^{-j4\omega} + 2(e^{-j\omega} - e^{-j3\omega})$

$= \left[(e^{j2\omega} - e^{-j2\omega}) + 2(e^{j\omega} - e^{-j\omega})\right] e^{-j2\omega}$

$= \underbrace{\left[2\sin(2\omega) + 4\sin(\omega)\right]}_{=A(\omega)} \cdot \underbrace{j}_{j = e^{j\pi/2}} \cdot \underbrace{e^{-j2\omega}}_{\text{linear phase } -2\omega}$

phase shift by $\pi/2$ (90°)

2. Zero Constellation Symmetry & System Synthesis

(a) Prove that if $z_0 = re^{j\theta}$ (with $r \neq 1$ and $\theta \neq 0, \pi$) is a zero of a causal, real-coefficient linear-phase FIR filter $H(z)$, then $z_0, z_0^*, 1/z_0$, and $1/z_0^*$ must also be zeros of $H(z)$ with equal multiplicity.

(b) A Type I FIR filter of order $M = 4$ ($N = 5$) with real impulse response $h[n]$ has a complex zero at:

$$z_1 = \frac{1}{2} e^{j\pi/3}$$

i. List all four zeros of $H(z)$ in both polar and Cartesian forms.

ii. Derive the normalized system transfer function $H(z)$ given that $H(e^{j0}) = 1$.

iii. Determine the explicit impulse response sequence $h[n]$ for $n = 0,1,2,3,4$.

(a) Use $H(z) = \pm z^{-M} \cdot H(z^{-1})$ for symmetric (+) or antisymmetric (-) FIR filters.

If $z_0 = re^{j\theta}$ (with $r>0$, $r \neq 1$ and $\theta \neq 0, \pi$) then

$H(z_0) = \pm (z_0)^{-M} \cdot H(z_0^{-1}) = 0$

$\Rightarrow$ Since $(z_0)^{-M} \neq 0$, if z_0 is a zero of $H(z)$, then $1/z_0$ must also be a zero of $H(z)$

For $H(z)$ with real-valued coefficients if z_0 is a zero of $H(z)$, then z_0^* and consequently $1/z_0^*$ also have to be zeros of $H(z)$.

Thus, the four zeros are

$$\{ z_0, 1/z_0, z_0^*, 1/z_0^* \}$$

In polar form

$$\left\{ re^{j\theta}, \frac{1}{r} e^{-j\theta}, re^{-j\theta}, \frac{1}{r} e^{j\theta} \right\}$$

(b) Type I FIR filter, $M = 4$ $(N = 5)$ with real impulse response has a zero at $z_1 = 0.5 e^{j\pi/3}$.

i. $z_1 = 0.5 e^{j\pi/3} = 0.25 + j0.433 = \frac{1}{4} + j\frac{\sqrt{3}}{4}$

$\frac{1}{z_1} = 2e^{-j\pi/3} = 1 - j1.732 = 1 - j\sqrt{3}$

$z_1^* = 0.5 e^{-j\pi/3} = 0.25 - j0.433 = \frac{1}{4} - j\frac{\sqrt{3}}{4}$

$\frac{1}{z_1^*} = 2 e^{j\pi/3} = 1 + j1.732 = 1 + j\sqrt{3}$

ii. $H(z) = (z - z_1)(z - \frac{1}{z_1})(z - z_1^*)(z - \frac{1}{z_1^*})$

$= \left(z^2 - \left(z_1 + \frac{1}{z_1}\right) z + 1\right)\left(z^2 - \underbrace{\left(z_1^* + \frac{1}{z_1^*}\right)}_{= \left(z_1 + \frac{1}{z_1}\right)^*} z + 1\right)$

$= z^4 - \left[\left(z_1 + \frac{1}{z_1}\right) + \left(z_1 + \frac{1}{z_1}\right)^*\right] z^3 + \left[2 + \left(z_1 + \frac{1}{z_1}\right)\left(z_1 + \frac{1}{z_1}\right)^*\right] z^2 +$

$- \left[\left(z_1 + \frac{1}{z_1}\right) + \left(z_1 + \frac{1}{z_1}\right)^*\right] z + 1$

$= z^4 - 2\underbrace{\mathrm{Re}\left\{z_1 + \frac{1}{z_1}\right\}}_{= \frac{5}{2}\cos\frac{\pi}{3} = \frac{5}{4}} z^3 + \left[2 + \underbrace{\left|z_1 + \frac{1}{z_1}\right|^2}_{= \left(\frac{5}{4}\right)^2 + \left(\frac{3\sqrt{3}}{4}\right)^2 = \frac{13}{4}}\right] z^2 - 2\underbrace{\mathrm{Re}\left\{z_1 + \frac{1}{z_1}\right\}}_{= \frac{5}{4}} z + 1$

$= z^4 - \frac{10}{4} z^3 + \frac{21}{4} z^2 - \frac{10}{4} z + 1$

Normalize: $H(1) = K \cdot \left(1 - \frac{10}{4} + \frac{21}{4} - \frac{10}{4} + 1\right) = K \cdot \frac{9}{4} = 1$

$\Rightarrow K = \frac{4}{9}$

$\Rightarrow H(z) = \frac{4}{9} z^4 - \frac{10}{9} z^3 + \frac{21}{9} z^2 - \frac{10}{9} z + \frac{4}{9}$

$= \underbrace{\left(\frac{4}{9} - \frac{10}{9} z^{-1} + \frac{21}{9} z^{-2} - \frac{10}{9} z^{-3} + \frac{4}{9} z^{-4}\right)}_{\text{causal } H(z)} z^4$

iii. $h[n] = \left[\frac{4}{9}, -\frac{10}{9}, \frac{21}{9}, -\frac{10}{9}, \frac{4}{9}\right]$, $n = 0, 1, 2, 3, 4$

3. System interconnections and composite types

Consider two causal linear-phase FIR filters: $H_A(z)$ of Type II with order M_A, and $H_B(z)$ of Type IV with order M_B.

(a) Determine the FIR Type (I, II, III, or IV) of the cascaded system $H_C(z) = H_A(z) \cdot H_B(z)$. Express the overall order M_C in terms of M_A and M_B, and justify the symmetry of $h_C[n]$.

(b) Assume $M_A = M_B = M$. Determine whether the parallel interconnection $H_D(z) = H_A(z) + H_B(z)$ maintains strict generalized linear phase. Support your answer mathematically by evaluating the phase of $H_D(e^{j\omega})$.

(c) Can a parallel combination of a Type I filter $H_1(z)$ and a Type III filter $H_3(z)$ with identical delay $\tau = M/2$ yield a linear phase system? Explain.

(a) Cascade system $H_c(z) = H_A(z) \cdot H_B(z)$:

Order $M_c = M_A + M_B$ (M_A: odd, M_B: odd) M_c is even order

Frequency response: $H_A(e^{j\omega}) = A_A(\omega)\, e^{-j\omega M_A/2}$ Type II

$H_B(e^{j\omega}) = j A_B(\omega)\, e^{-j\omega M_B/2}$ Type IV

$$H_c(e^{j\omega}) = H_A(e^{j\omega}) \cdot H_B(e^{j\omega})$$

$$= j A_A(\omega) A_B(\omega)\, e^{-j\omega (M_A+M_B)/2}$$

$$= j A_c(\omega)\, e^{-j\omega M_c/2}$$

Since M_c is even and $H_c(j\omega)$ contains the factor $j = e^{j\pi/2}$ (anti-symmetric impulse response $h_c[n] = -h_c[M_c - n]$), $H_c(z)$ is a Type III IIR filter.

(b) Parallel system: $H_D(z) = H_A(z) + H_B(z)$

with $M_A = M_B = M$: (H_A: Type II, H_B: Type IV)

$$H_D(e^{j\omega}) = A_A(\omega)\, e^{-j\omega M/2} + jA_B(\omega)\, e^{-j\omega M/2}$$

$$= [A_A(\omega) + jA_B(\omega)]\, e^{-j\omega M/2}$$

Let $A_D(\omega) = A_A(\omega) + jA_B(\omega) = |A_D(\omega)|\, e^{j\theta(\omega)}$, where

$$\theta(\omega) = \tan^{-1}\left(\frac{A_B(\omega)}{A_A(\omega)}\right)$$

The total phase response is

$$\angle H_D(e^{j\omega}) = \theta(\omega) - \frac{\omega M}{2}$$

But $\theta(\omega)$ is a non-linear function of ω

$\Rightarrow$ The parallel combination does not have strict linear phase in general.

(c) $$H(e^{j\omega}) = H_1(e^{j\omega}) + H_3(e^{j\omega}) = [A_1(\omega) + jA_3(\omega)]\, e^{-j\omega M/2}$$

$$= |A_1(\omega) + jA_3(\omega)|\, e^{j\theta(\omega)}\, e^{-j\omega M/2}$$

with $$\theta(\omega) = \tan^{-1}\left(\frac{A_3(\omega)}{A_1(\omega)}\right)$$

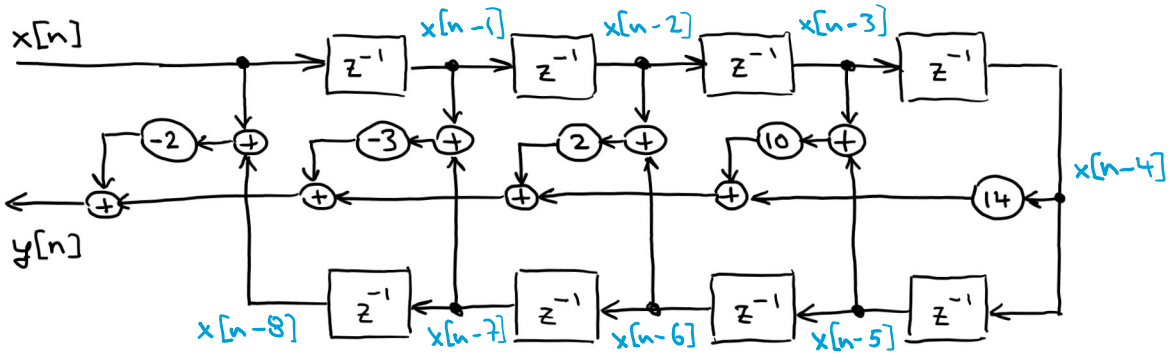
As in part (b) the total phase is

$$\angle H(e^{j\omega}) = \theta(\omega) - \frac{\omega M}{2}$$

Since $\theta(\omega)$ is non-linear in ω in general the parallel combination of a Type I and a Type III will not have linear phase in general.

4. Filter flowgraph (ECEN 5632 Problem)

Consider the filter flowgraph shown in the figure below:



The z^{-1} cells are delay cells and the bubbles with numbers in them are multipliers with the numbers giving the multiplication factors. The bubbles with + signs are adders, the branch labeled $x[n]$ is the input of the filter and the branch labeled $y[n]$ is the output of the filter.

(a) Determine the unit impulse response $h[n]$ of the filter shown above and display the result in the form of a labeled stem plot. What type of a filter is it?

(b) What would be the main advantage of implementing a filter in this way? Are there any disadvantages?

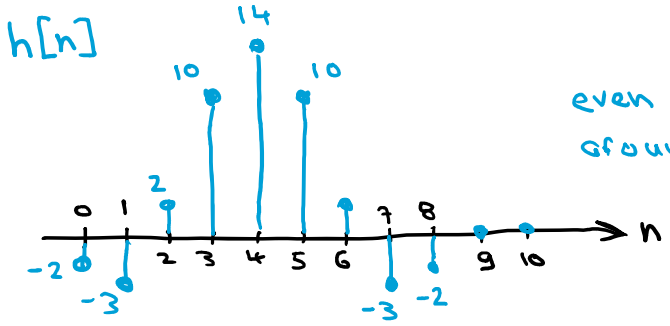
(c) Could you implement a Type III and/or Type IV FIR filter in this way? If yes, what would you have to change, if no, why not?

(a) $y[n] = -2x[n] - 3x[n-1] + 2x[n-2] + 10x[n-3] +$
$+ 14x[n-4] + 10x[n-5] + 2x[n-6] - 3x[n-7] +$
$- 2x[n-8]$

Setting $x[n] = \delta[n]$ yields the impulse response

$h[n] = [-2, -3, 2, 10, 14, 10, 2, -3, -2]$

FIR Type I
order $M = 8$



even symmetry around $n = 4$

(b) Advantages:

- It exploits $h[n] = h[M-n]$ by adding two samples that have the same coefficient $\Rightarrow$ numbers of multiplications goes from $M+1$ to $M/2+1$
- Linear phase is guaranteed, even after coefficient quantization

Disadvantages:

- Number of adders and delays is not reduced
- The structure only works for filters with exact (anti) symmetry

In most practical cases the advantages would probably outweigh the disadvantages.

(c) Yes you could easily implement a Type III or IV filter this way. You would just use subtractors instead of adders before the coefficient multiplication and no middle tap multiplier fo Type III.